

Software Architecture & System Design Document (SDA)

1. Executive Summary

This document defines the system design, front-end architecture, and design specifications for **Thirueesvaran Thisanthan's Portfolio Application**. The portfolio serves as a personal brand showcase, featuring personal bio, educational background, technical stack, key projects, and contact capability.

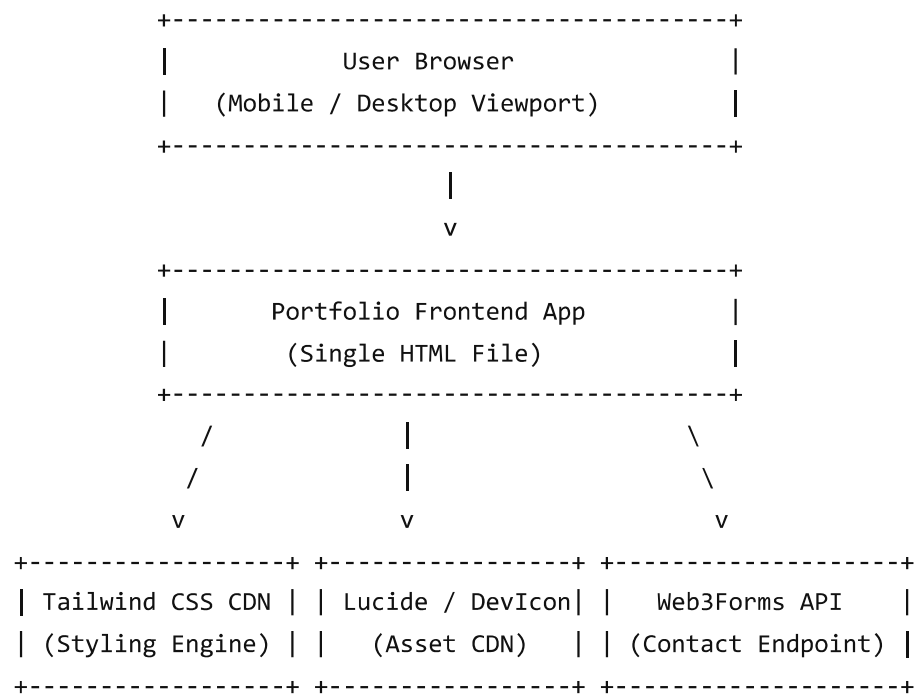
2. System Overview & Objectives

- **System Purpose:** Showcase software development, web development, and UI/UX design capabilities.
- **Architecture Style:** Single Page Application (SPA) / Single-File Static Web Application.
- **Deployment Target:** Web browsers (responsive across mobile, tablet, and desktop).

3. Technology Stack & External Dependencies

Module / Layer	Technology / Source	Description
Markup & Styling	HTML5, Tailwind CSS (v3 CDN)	Core structure and modern utility-first CSS styling
Icons & Media	Lucide Icons, DevIcon CDN	Scalable vector icons for interface elements and tech badges
Form Processing	Web3Forms API (https://api.web3forms.com/submit)	Serverless contact form handling
Assets	Local Images/Videos (profile.jpeg, project-1.mp4, etc.)	Portfolio imagery and video demonstrations

4. High-Level System Architecture



5. UI/UX Design System & Color Palette

- **Brand Primary:** #FF5500
- **Brand Hover:** #E04800
- **Dark Background (bg):** #0B0B0E
- **Dark Card (card):** #13131A
- **Dark Border (border):** #22222E

6. Functional Module Decomposition

- **Navigation Header (#navbar):** Fixed navigation, branding, section scroll targets, and contact action link.
- **Hero Module (#home):** Candidate introduction, action buttons, dynamic photo container, and code snippet card.
- **Profile & Bio (#about):** Metadata widgets (Hometown, Age, University, Focus) and short description.
- **Academic Background (#education):** Undergraduate degree details, modules, and certificate link.
- **Skills & Tech Stack (#skills):** Badges for Java, C, C++, OOP, Web Tech, and UI/UX.

- **Portfolio Showcase (#projects):** Projects including Java App, Blood Donation System, and UI/UX videos.
- **Communication & Contact (#contact):** Web3Forms submission form and personal contact metadata.